

Comparing Models

To compare function models, you must identify key characteristics of the various models.

Using Multiple Representations

Function models can be represented algebraically, graphically, numerically, or verbally.

EXAMPLE 1

Give algebraic, graphic, and numerical representations of the verbal model:
At Mug Manufacturing Inc., the total cost of production is based on a fixed overhead cost of \$20 and a unit cost of \$3/mug.

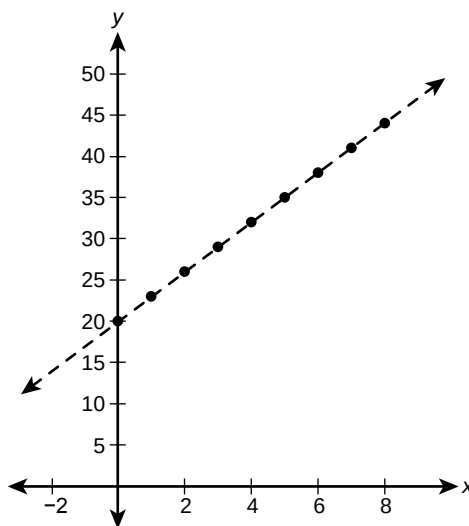
SOLUTION

The situation has a fixed unit cost, so a linear model makes sense. Specifically, the fixed overhead cost can be modeled with the y -intercept, and the unit cost can be modeled with the slope. An algebraic representation is $f(x) = 3x + 20$.

The model is a line with slope 3 and y -intercept 20, and the domain is nonnegative whole numbers. A graphical representation is shown.

A numerical representation is shown in the table.

x	0	1	2	3	5	10	20
$f(x)$	20	23	26	29	35	50	80



Describing Functions

All functions can be described by various characteristics. These characteristics focus on the relationship of the input and associated output of the function.

Characteristics of Function Models

A function model can be described with these characteristics:

- domain and range
- zeros
- x- and y-intercepts
- minimum and maximum values
- intervals of positive and negative values
- increasing, decreasing, and constant intervals
- lines of symmetry
- even and odd behavior
- rate of change
- amplitude and period

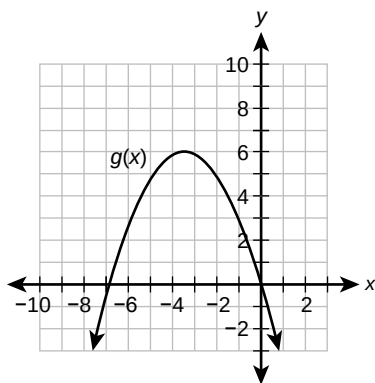
Comparing Function Models

When comparing function models, use characteristics that apply to the models being compared.

EXAMPLE 2

Model A and Model B are two quadratic models. Model A is represented graphically, and Model B is represented algebraically. Compare the two models.

Model A



Model B

$$f(x) = 1 - x^2 + 4x$$

SOLUTION

For these models, compare domain and range, zeros, absolute maximum and minimum values, and intervals for which the models are positive and negative.

Because Model A is easier to interpret, focus on Model B.

Write $f(x) = 1 - x^2 + 4x$ as $f(x) = -(x - 2)^2 + 5$ to find the vertex, which is the point $(2, 5)$.

Use the quadratic formula to find the zeros.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-4 \pm \sqrt{4^2 - 4(-1)(1)}}{2(-1)} = \frac{-4 \pm \sqrt{20}}{-2} = \frac{-4 \pm 2\sqrt{5}}{-2} = 2 \pm \sqrt{5}$$

Compare the models using a table.

Characteristic	Model A	Model B
domain	all real numbers	all real numbers
range	$g(x) \leq 6$	$f(x) \leq 5$
zeros	$x = 0$ and $x = -7$	$x = 2 \pm \sqrt{5}$
absolute maximum and minimum values	maximum value of $y = 6$	maximum value of $y = 5$
intervals of positive and negative values	positive values: $-7 < x < 0$ negative values: $x < -7$ and $x > 0$	positive values: $2 - \sqrt{5} < x < 2 + \sqrt{5}$ negative values $x < 2 - \sqrt{5}$ and $x > 2 + \sqrt{5}$